

BHARATH PRIYAN K

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<https://bharathoo7.github.io/portfolio/> | [LinkedIn](#) | [GitHub](#)



SUMMARY

Analytical and performance-driven Software Developer with a B.Tech in Computer Science (AI & ML). Proficient in Python, C++, and SQL, with a strong foundation in developing scalable, high-efficiency applications. Experienced in the full software development life cycle, from designing low-latency ML systems to deploying robust models using MLOps. Adept at writing testable, clean code to solve complex problems in mission-critical environments

TECHNICAL SKILL

Programming: Python, C++, SQL, Shell Scripting.

Application Development: SDLC Concepts, Object-Oriented Programming (OOPS), Data Structures & Algorithms (DSA), Test-Driven Development (TDD).

Enterprise Tools & DevOps: Git, CI/CD, Docker, GitHub Actions, AWS (S3, SageMaker), Azure.

Web Technologies: FastAPI, Streamlit, React.js, Node.js.

AI & Data: TensorFlow, PyTorch, LangChain, FAISS, Pandas, NumPy, Power BI.

Soft Skills: Communication, Teamwork, Leadership, Critical Thinking, Adaptability, Rapport Building, Stakeholder Management, People Management.

PROFESSIONAL EXPERIENCE

UPSC Civil Services Preparation (Professional Sabbatical) Self-Directed Study **June 2024 – Present**

- Dedicated full-time to intensive preparation for the Civil Services Examination, focusing on governance, ethics, essential for the Responsible Use of AI and compliance with internal policies and regulations.
- Developed high-level discipline, time management, and a macro-perspective on systemic problem-solving and practiced making well-grounded decisions based on vast amounts of data and insights.

AI-ML Virtual Internship (Remote) **May 2023 - July 2023**

- Preprocessed a loan prediction dataset with 10,000+ entries using Pandas and NumPy, improving model training efficiency by 25%.
- Improved model accuracy by 15% using Random Forest, deployed and tested in Google Colab.

EDUCATION

SRM Institute of Science and Technology **2020 - 2024**

Bachelor of Technology in Computer Science with specialization in (AI & ML) | CGPA: 8.72

KEY PROJECTS

MedBot AI (RAG Chatbot) | [GitHub](#)

- Developed a Retrieval-Augmented Generation (RAG) chatbot using LangChain, Hugging Face LLMs, and a FAISS vector store for medical queries.
- Built a Streamlit UI for real-time semantic search and answer generation from a private document corpus.

Adaptive Traffic Signal Control System | [GitHub](#)

- Implemented a Deep Reinforcement Learning (DRL) agent in Python to automate and optimize traffic flow, demonstrating a 30% reduction in average wait time in a simulated environment.

Tire Condition Detection & Segmentation | [GitHub](#)

- Engineered a PyTorch and OpenCV computer vision model to detect and segment tire wear, achieving over 90% pixel accuracy on a custom dataset.

ADDITIONAL INFORMATION

Certifications:

- Full-Stack Web Development with React – Coursera
- MATLAB: Deep Learning, Machine Learning, Data Visualization, AWS OnRamp
- ISRO – Applications of ML in Urban Studies
- DevOps & Node.js/React.js Workshops – 2023

Languages: English (Fluent), Tamil (Native), Telugu (Conversational)

Interests: Fitness, Automobiles, Coding, Trekking